* The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 January 1997. *

* INCH POUND *

MIL-PRF-19500/635
16 October 1996

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
ULTRAFAST, TYPES 1N6710 THROUGH 1N6716,
JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, ultrafast power rectifier diodes. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

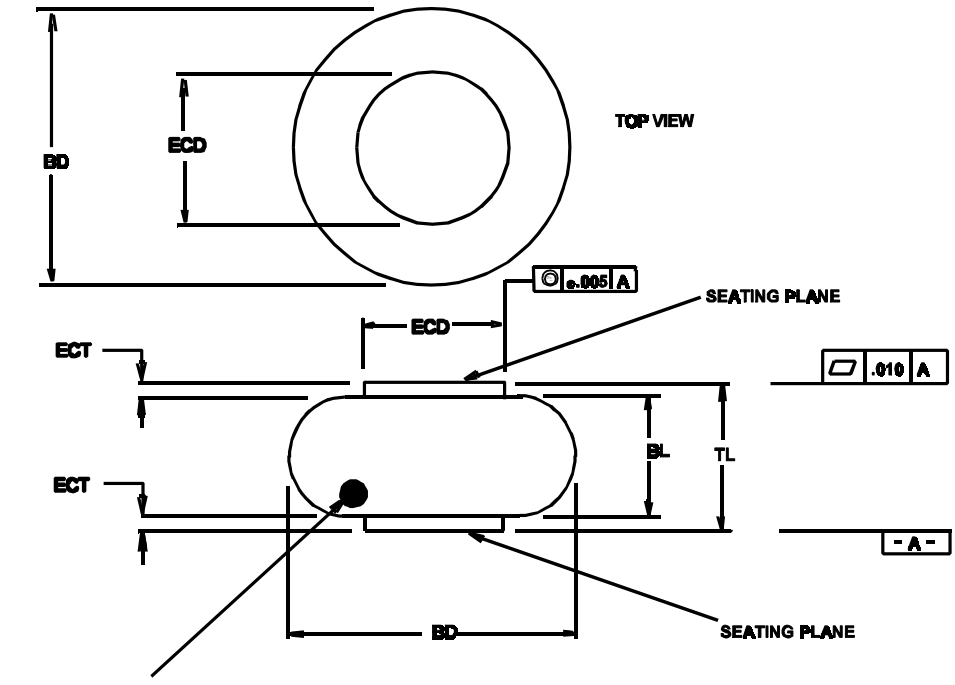
Types	V_{RWM}	I_O 1/	t_{rr}	I_{FSM}	$R_{\Theta JE}$	T_{STG} and T_J
		T_A = +25°C		t_p = 8.3 ms, T_A = +25°C		
	V dc	A dc	ns	A (pk)	°C/W	
1N6710	100	50	40	375	1.2	-65 to +175
1N6711	200	50	40	375	1.2	-65 to +175
1N6712	400	50	40	375	1.2	-65 to +175
1N6713	600	50	60	375	1.2	-65 to +175
1N6714	800	50	60	375	1.2	-65 to +175
1N6715	1,000	40	80	300	1.2	-65 to +175
1N6716	1,200	40	80	300	1.2	-65 to +175

1/ Derate linearly 1N6710-1N6712 at 625 mA/°C from +95°C to +175°C.
Derate linearly 1N6713-1N6714 at 417 mA/°C from +55°C to +175°C.
Derate linearly 1N6715-1N6716 at 267 mA/°C from +25°C to +175°C.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. *

AMSC N/A FSC 5961
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.



Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
ECD	0.215	0.235	5.46	5.97
BL	0.110 TYP		2.79 TYP	
TL	0.120	0.150	3.048	3.81
BD		0.370		9.398
ECT	0.005		0.127	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Dimensions and configuration for 1N6710 through 1N6716 (D0-217AB).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Services Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 herein. The US government's preferred system of measurement is the metric system. However, since this item was originally designed using inch-pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

3.3.1 Diode construction. These devices shall be constructed utilizing non-cavity double plug construction with metallurgical bonding between both sides of the silicon die and terminal pins (see MIL-PRF-19500). Metallurgical bond shall be in accordance with the requirements of category 1 in MIL-PRF-19500.

3.3.2 Lead formation and finish. Lead finish shall be solderable as defined in MIL-PRF-19500 and herein. Where a choice of lead finish or formation is desired, it shall be specified in the acquisition document (see 6.2).

3.3.3 Polarity. Polarity shall be marked with three dots, 120° apart on the cathode side of the diode body.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500 and herein. All marking may be omitted from the device except for the cathode marking. All marking which is omitted from the body of the device shall appear on the label of the initial container.

3.5 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.3 Screening. Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c 1/	Thermal impedance (see 4.5.2)	Thermal impedance (see 4.5.2)
9	Not applicable	Not applicable
10	Not applicable	Not applicable
11	V_{F1} and I_{R1}	V_{F1} and I_{R1}
12	MIL-STD-750, method 1038; test condition A; t = 240 hours V_R = 80 percent of rated V_R; T_A = +150°C	MIL-STD-750, method 1038; test condition A; t = 48 hours V_R = 80 percent of rated V_R; T_A = +150°C
13 2/ 3/	Subgroup 2 and 7 of table I herein; V_{F1} and I_{R1}; ΔV_{F1} = ±150 mV (pk); ΔI_{R1} = ±5 µA dc or 100 percent from the initial value, whichever is greater.	Subgroup 2 and 7 of table I herein; V_{F1} and I_{R1}; ΔV_{F1} = ±150 mV (pk); ΔI_{R1} = ±5 µA dc or 100 percent from the initial value, whichever is greater.

1/ Thermal impedance shall be performed any time after screen 3.
2/ Thermal impedance does not need to be repeated.
3/ Delta readings shall be performed within 16 hours after removal of applied voltage in HTRB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. The following test conditions shall be used for $Z_{\Theta JX}$, group A inspection:

a. I_M measure current - - - - - - - - - - - - - - - - - 10 mA minimum

b. I_H forward heating current - - - - - - - - - - - - 30 A minimum

c. t_H heating time - 10 ms minimum

d. t_d measurement delay time - - - - - - - - - - 100 µs minimum

The maximum limit for $Z_{\Theta JX}$ under these test conditions are $Z_{\Theta JX}$(max) = .5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, group A, subgroup 2 (except thermal impedance).

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals. Condition A, T_A = +25°C.
4	1037	ΔT_C = 85°C minimum, I_F or I_O = 2 A minimum for 2,000 cycles; endpoints per 4.4.5 herein.
5	1027	I_F = 2 A dc minimum, adjust T_A or I_F to achieve T_J = 200°C, +0°C, -35°C, t = 240 hours min; (heatsinking allowed) endpoints per 4.4.5 herein.
6	3101	Not applicable.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
2	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals. Condition B, T_A = +25°C.
3	1037	ΔT_C = 85°C minimum, I_F or I_O = 2 A minimum for 2,000 cycles, endpoints per 4.4.5 herein.
5	3101	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, group A, subgroup 2 (except thermal impedance).

Subgroup	Method	Condition
2 and 3	----	Not applicable.
6	1037	ΔT_C = 85°C minimum, I_F or I_O = 2 A minimum for 6,000 cycles, endpoints per 4.4.5 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, group A, subgroup 2 (except thermal impedance) herein and 4.4.5.

4.4.4.1 Group E inspection, table IX of MIL-PRF-19500.

Subgroup	Method	Condition	Sampling plan
1	1051	Condition C, 500 cycles.	22 devices, c = 0
2	1038	Condition A, t = 1,000 hours V_R = 80 percent V_{rated}, T_A = +125°C	22 devices, c = 0
3		Not applicable	
4	3101	T_A = +25°C; $R_{\Theta JEC}$ = rated $R_{\Theta JEC}$ (see 1.3).	5 devices, c = 0
5	1001	Not applicable	

4.4.5 Group B, C, and E electrical measurements (delta requirements).

Test	Method	Condition	Symbol	Maximum	Unit
V_F	4011	I_F = 5 A (pk); pulsed (see 4.5.1)	V_{F1}	per table 1	V
ΔV_F	4011		ΔV_{F1}	±150	mV
I_R	4016	DC method, V_R = rated V_R (see 1.3); pulsed (see 4.5.1)	I_{R1}	50	µA
ΔI_R	4016		ΔI_{R1}	5µA or 100 percent, whichever is greater.	µA

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in 4.3.2.1 of MIL-STD-750.

4.5.2 Thermal impedance $Z_{\Theta JX}$ measurements for screening. The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\Theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition.

4.5.2.1 Thermal impedance ($Z_{\Theta JX}$ measurements) for initial qualification or requalification. The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record date $Z_{\Theta JX}$). $Z_{\Theta JX}$ shall be supplied on one lot (500 pieces minimum) and a thermal response curve shall be submitted. Twenty two of these samples shall be serialized and provided to the qualifying activity for correlation prior to shipment of parts. Measurements conditions shall be in accordance with 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\Theta JX}$		0.5	°C/W
Breakdown voltage	4022	I_R = 500 µA dc	V_{BR}			
1N6710				110		V
1N6711				210		V
1N6712				410		V
1N6713				620		V
1N6714				820		V
1N6715				1020		V
1N6716				1220		V
Forward voltage	4011	I_F = 5 A (pk); pulsed (see 4.5.1)	V_{F1}			
1N6710 - 1N6712					0.8	V
1N6713, 1N6714					1.0	V
1N6715, 1N6716					1.25	V
Forward voltage	4011	I_F = 50 A (pk); pulsed (see 4.5.1)	V_{F2}			
1N6710 - 1N6712					1.1	V
1N6713, 1N6714					1.6	V
1N6715, 1N6716					3.0	V
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R1}		50	µA
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R2}		5.0	mA
Low temperature operation:		T_A = -55°C				
Forward voltage	4011	I_F = 5 A (pk); pulsed (see 4.5.1)	V_{F3}			
1N6710 - 1N6712					0.9	V
1N6713, 1N6714					1.1	V
1N6715, 1N6716					1.35	V

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 4						
Reverse recovery time	4031	Condition B; I_F = .5 A, I_{RM} = 1 A $I_{R(REC)}$ = .25 A	t_{rr}			
1N6710 - 1N6712					40	ns
1N6713, 1N6714					60	ns
1N6715, 1N6716					80	ns
Junction capacitance	4001	V_R = 10 V dc, f = 1 MHz V_{SIG} = 50 mV (p-p) (max)	C_J			
1N6710 - 1N6712					750	pF
1N6713, 1N6714					550	pF
1N6715, 1N6716					250	pF
Subgroups 5 and 6						
Not applicable						
Subgroup 7						
Scope display evaluation	4023 2/					

1/ For sampling plan, see MIL-PRF-19500.

2/ The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 20 to 50 µA/division and 20 to 200V/division. Reverse current over the knee shall be at least 100 µA. Each device may exhibit a slightly rounded characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agencys' automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Preparing activity:
DLA - ES

(Project 5961-1848)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/635	**2. DOCUMENT DATE (YYMMDD)** 96/10/16

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, ULTRAFAST, TYPES 1N6710 THROUGH 1N6716, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0516
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T04:24Z
Check the source to verify that this is the current version before use.